

Drew Gaddy

Procambarus clarkii (Girard, 1852)

Common name: Red Swamp Crayfish

Synonyms and Other Names: Red swamp crayfish/crawfish, Louisiana crayfish/crawfish, *Cambarus clarkii* Girard, 1852

Taxonomy: available through ITIS

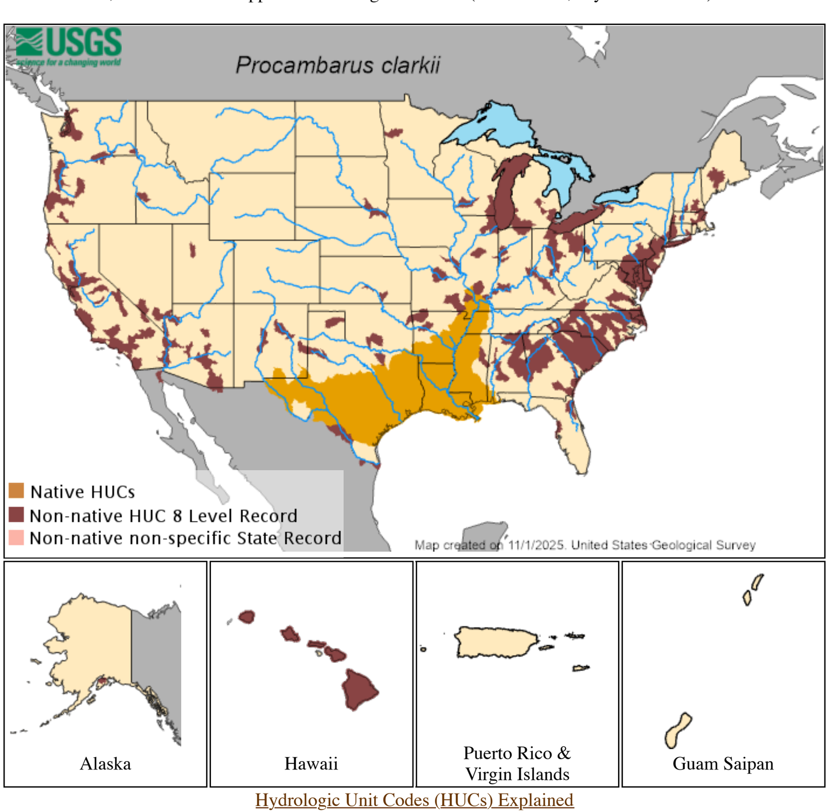
Identification: The red swamp crayfish is typically dark red, with elongate claws (chelae) and head, a triangular rostrum tapering anteriorly without a central keel, reduced or absent spines on the side of the shell (carapace) between the head and thorax, and a linear to obliterate dorsal surface between the 2 carapace plates (areola), which converge (Boets et al. 2009, GISD 2011, NatureServe 2011). The first walking leg (cheliped) bears bright red rows of bumps (tubercles) on its side (mesial) margin and palm.

In reproductively mature males, hooks are present on the third segment (from the base; the ischia) of the third and fourth pairs of walking legs, and the first swimmeret (pleopod) of ends in four projections (terminal elements), with the most anterior terminal end (cephalic process) of this sperm transfer structure rounded with a sharp angle on the outer (caudodistal) margin, which lacks "hairs" (setae) below its tip. Setae on the anterior surface of the pleopod, closest to the terminal elements, have strong angular shoulders. The right pleopod is wrapped around the side, such that it appears reduced or absent, and possesses a spur on the inner margin on its fifth joint (carpopodite) (WDFW 2003). Strong spines project from the inner face of the sixth joint (propodite); "knots" are present on the dorsal face or this joint (Boets et al. 2009).

Juveniles are not red and are difficult to distinguish from other *Procambarus* species (Boets et al. 2009).

Size: Adults range in length from 5.5 to 12 centimeters (or 2.2 to 4.7 inches) and may attain weights in excess of 50 grams in 3 to 5 months (GIS 2011, Hentonnen and Huner 1999).

Native Range: Gulf coastal plain from the Florida panhandle to Mexico; southern Mississippi River drainage to Illinois (Hobbs 1989, Taylor et al. 2007).



Hydrologic Unit Codes (HUCs) Explained
Interactive maps: Point Distribution Maps

Nonindigenous Occurrences:

Table 1. States with nonindigenous occurrences, the earliest and latest observations in each state, and the tally and names of HUCs with observations†. Names and dates are hyperlinked to their relevant specimen records. The list of references for all nonindigenous occurrences of *Procambarus clarkii* are found here.

State	First Observed	Last Observed	Total HUCs with observations†	HUCs with observations†
AL	1992	2021	11	Guntersville Lake; Lower Black Warrior; Lower Coosa; Lower Tallapoosa; Middle Alabama; Middle Coosa; Middle Tallapoosa; Middle Tennessee-Chickamauga; Sipsey; Upper Coosa; Wheeler Lake
AK	2004	2004	1	Upper Kenai Peninsula
AZ	1981	2024	12	Bill Williams; Canyon Diablo; Hassayampa; Havasu-Mohave Lakes; Imperial Reservoir; Lower Colorado; Lower Salt; Middle Gila; Rillito; Tonto; Upper San Pedro; Upper Santa Cruz
AR	2004	2021	2	Dardanelle Reservoir; Illinois
CA	1924	2024	41	Aliso-San Onofre; California; California Region; Calleguas; Central California Coastal; Central Coastal; Clear Creek-Sacramento River; Coyote; Crowley Lake; Fresno River; Imperial Reservoir; Los Angeles; Lower American; Lower Colorado; Lower Sacramento; Mad-Redwood; Middle Kern-Upper Tehachapi-Grapevine; Mojave; Owens Lake; Russian; Sacramento-Stone Corral; Salinas; San Diego; San Francisco Bay; San Gabriel; San Joaquin; San Luis Rey-Escondido; San Pablo Bay; Santa Ana; Santa Ana; Santa Barbara Coastal; Santa Clara; Santa Monica Bay; Santa Ynez; Southern California Coastal; Suisun Bay; Tulare-Buena Vista Lakes; Upper Coon-Upper Auburn; Upper Cosumnes; Upper Mokelumne; Upper Tuolumne
CT	2009	2023	4	Housatonic; Lower Hudson; Saugatuck; Thames
DE	2013	2019	3	Brandywine-Christina; Broadkill-Smyrna; Chincoteague
FL	1951	2024	2	Crystal-Pithlachascotee; Lower St. Johns
GA	1989	2024	13	Cumberland-St. Simons; Etowah; Middle Chattahoochee-Lake Harding; Middle Chattahoochee-Walter F; Middle Savannah; Ogeechee Coastal; Oostanaula; Satilla; Upper Chattahoochee; Upper Coosa; Upper Flint; Upper Ocmulgee; Upper Oconee
HI	1923	2019	6	Hawaii; Hawaii Region; Kauai; Maui; Molokai; Oahu
ID	1975	2024	3	Lower Boise; Lower Snake-Asotin; Pacific Northwest Region
IL	1973	2024	9	Big Muddy; Chicago; Des Plaines; Lake Michigan; Lower Fox; Lower Ohio-Bay; Salt; The Sny; Upper Fox
IN	1989	2021	4	Highland-Pigeon; Lower Ohio-Little Pigeon; Middle Wabash-Little Vermilion; Ohio Region
KS	2019	2019	2	Gar-Peace; Lower Walnut River
KY	2013	2013	1	Licking
ME	1980	2003	2	Lower Kennebec River; New England Region
MD	1989	2024	16	Chester-Sassafras; Chincoteague; Choptank; Gunpowder-Patapsco; Lower Potomac; Lower Susquehanna; Mid Atlantic Region; Middle Potomac-Anacostia-Occoquan; Middle Potomac-Catoctin; Monocacy; Nanticoke; Patuxent; Pokomoke-Western Lower Delmarva; Severn; Tangier; Upper Chesapeake Bay
MA	2021	2024	4	Ashuelot River-Connecticut River; Blackstone River; Charles; Merrimack River
MI	2013	2025	6	Black-Macatawa; Clinton; Detroit; Shiawassee; Southeastern Lake Michigan; St. Joseph
MN	2016	2016	1	Eastern Wild Rice
MS	2015	2015	2	Noxubee; Tibbee
MO	1985	2025	7	Beaver Reservoir; Cahokia-Joachim; Current; Lower Missouri-Moreau; Meramec; Upper Black; Whitewater
NE	2016	2016	1	Lewis and Clark Lake
NV	1942	2018	3	Lake Mead; Las Vegas Wash; Upper Amargosa
NJ	2016	2022	7	Cohansey-Maurice; Great Egg Harbor; Hackensack-Passaic; Lower Delaware; Middle Delaware-Musconetcong; Raritan; Sandy Hook-Staten Island
NM	1965	2013	2	Pecos Headwaters; Rio Grande-Albuquerque
NY	2002	2025	5	Lower Hudson; Niagara River; Northern Long Island; Sandy Hook-Staten Island; Southern Long Island
NC	1989	2023	31	Albemarle; Cape Fear; Chowan; Coastal Carolina; Fishing; Haw; Lower Catawba; Lower Pee Dee; Lower Pee Dee; Lower Roanoke; Lower Tar; Lower Yadkin; Lumber; Meherrin; Middle Neuse; Neuse; Northeast Cape Fear; Pamlico; Rocky; Tuckasegee; Upper Broad; Upper Cape Fear; Upper Catawba; Upper French Broad; Upper Neuse; Upper Pee Dee; Upper Pee Dee; Upper Tar; Upper Yadkin; Waccamaw; White Oak River
OH	1967	2024	11	Black-Rocky; Cedar-Portage; Cuyahoga; Lake Erie; Licking; Lower Scioto; Mahoning; Middle Ohio-Laughery; Sandusky; Tuscarawas; Upper Scioto
OK	1989	2017	3	Arkansas-White-Red Region; Middle Washita; Northern Beaver
OR	1989	2024	12	Coos; Coquille; Middle Columbia-Lake Wallula; Middle Rogue; Middle Willamette; North Umpqua; Pacific Northwest Region; Tualatin; Umpqua; Upper Deschutes; Upper Willamette; Yamhill
PA	1990	2025	7	Crosswicks-Neshaminy; Lower Delaware; Lower Susquehanna; Lower West Branch Susquehanna; Middle Delaware-Musconetcong; Schuylkill; Upper Ohio
SC	1988	2023	29	Black; Carolina Coastal-Sampit; Coastal Carolina; Congaree; Cooper; Edisto River; Enoree; Four Hole Swamp; Lake Marion; Little Pee Dee; Lower Broad; Lower Catawba; Lower Pee Dee; Lower Savannah; Lumber; Lynches; North Fork Edisto; Salkehatchie; Saluda; Santee; Seneca; South Atlantic-Gulf Region; South Carolina Coastal; South Fork Edisto; Stevens; Upper Broad; Upper Savannah; Waccamaw; Wateree
TX	1985	2020	6	Elm Fork Trinity; Lower Rio Grande; Middle Canadian-Spring; North Fork Double Mountain Fork Brazos; San Ambrosia-Santa Isabel; Yellow House Draw
UT	1978	1989	1	Rush-Tooele Valleys
VA	1999	2022	7	Albemarle; Appomattox; Lower Chesapeake; Lower James; Middle James-Willis; Middle Potomac-Anacostia-Occoquan; Upper New
WA	2000	2024	5	Duwamish; Lake Washington; Lower Skagit; Puget Sound; Puyallup
WI	2009	2019	3	Lower Wisconsin; Milwaukee; Pike-Root

Table last updated 12/3/2025

† Populations may not be currently present.

Ecology: This species lives in a variety of freshwater habitats, including rivers, lakes, ponds, streams, canals, seasonally flooded swamps and marshes, and ditches with mud or sand bottoms and plenty of organic debris (Huner and Barr 1991). *Procambarus clarkii* also frequently colonizes rice fields, irrigation channels, and reservoirs (Correia and Ferreira 1995, Gherardi et al. 1999). It exhibits considerable ecological plasticity and is tolerant of a range of salinities up to 35ppt (Bissattini et al. 2015), (2-3 ppt for reproduction), pH (5.8-10), oxygen levels (>3 ppm), temperatures (as long as water in burrows neither freezes nor exceeds 35°C), and pollution levels (Huner and Barr 1991). Although this species is known to have a preference for habitats with water temperature from 21 to 30 °C, (Peruzza et al. 2015) demonstrated that the red swamp crayfish could adapt to atypical thermal habitat, characterized by an annual mean water temperature values of 13 °C. Studies of the red swamp crayfish invasion in Europe suggest that it tends to prefer areas of lower flow velocity and low elevation; in central and southern Europe, it has established in warm, shallow natural and agricultural wetlands while in northern Europe, it can be found in small permanent ponds free of fish predation (Cruz and Rebelo 2007, Henttonen and Huner 1999).

The red swamp crayfish is a physical ecosystem engineer, primarily constructing simple, two-crayfish burrows consisting of a single opening, which may be covered with a mud plug or chimney to reduce evaporative loss further from the water's edge, and a tunnel widening to an enlarged terminal chamber (Correia and Ferreira 1995, Huner and Barr 1991, Jaspers and Avault 1969). In periods of drought or elevated temperatures, these burrows can extend 40-90 cm down to water table (Ingle 1997). Burrow density is typically greatest in areas with fine sediments and lowest in areas of sand, gravel, or cobble (Barbaresi et al. 2004). Where present, Myriophyllum sp., fallen logs, and other vegetation may encourage greater burrow density (Correia and Ferreira 1995). Water hyacinth (*Eichhornia crassipes*) has also provided habitat for this crayfish in other introduced populations (Smart et al. 2002).

Like most crayfish, the red swamp crayfish is an opportunistic omnivore, consuming plant material, animals, detritus, and sediment (Alcorlo et al. 2004; Anastácio et al. 2005; Correia 2003; Gherardi and Barbaresi 2007, 2008; Gutiérrez-Yurrita et al. 1998; Hobbs 1993; Ilheu and Bernardo 1993; Pérez-Bote 2004; Smart et al. 2002). In terms of feeding preference, a few trends have emerged from studies of native and introduced populations. Plants and/or detritus tend to be consumed in greatest frequency and volume, with plant consumption highest in summer and detritus feeding intense year round (Correia 2003, Gherardi and Barbaresi 2008). It appears that crayfish may exhibit selectivity for particular plants but not among animal prey (Gherardi and Barbaresi 2007). The animal constituents of the red swamp crayfish diet tend to be dominated by insects (particularly chironomids), other crayfish, mollusks (snails), and fish (Ilheu and Bernardo 1993, Pérez-Bote 2004). Juveniles consume more animals than adults, which exhibit an ontogenic shift in diet to plants and detritus, but cannibalism is most apparent in adults and preadults (Correia 2003, Pérez-Bote 2004). Fish is also an important staple of the adult winter diet, and males may eat fish in a higher proportion than do females. This may be attributed to large claw size in some males and potentially also due to higher male mobility during the mating season (Ilheu and Bernardo 1993, Pérez-Bote 2004). However, the nutritional benefit of carnivory may be outweighed by the cost of active predation, leading to increased herbivory or detritivory in the field (Ilheu and Bernardo 1993). Overall consumption is highest in the fall and winter (Pérez-Bote 2004).

The life cycle of the red swamp crayfish is relatively short, with an onset of sexual maturity occurring in as few as two months and a total generation time of four and a half months (Huner and Barr 1991). Breeding typically taking place in the fall, though in warmer, wetter regions, there may be a second reproductive period in the spring. This species exhibits high fecundity: a 10 cm female can produce as many as 500 eggs, while a smaller female produces around 100 eggs (GISD 2011, Huner and Barr 1991). Egg production make take as short a period as six weeks, followed by a three-week period of incubation and maternal attachment and an additional eight weeks until egg maturation (GISD 2011). *Procambarus clarkii* females incubating eggs or carrying young may be found year-round, which contributes greatly to the success and abundance of this species, but optimal temperatures are 21-27°C; growth is inhibited below 12°C (Ackefors 1999, GISD 2011). Recently hatched crayfish remain in the burrow with their mother as long as eight weeks and must molt twice before being self-sufficient (Hunter and Barr 1991). Due to the cannibalistic nature of conspecifics in communal burrows, adult molting often occurs in the open, even in the presence of predatory fish (Hartman and O'Neill 1999). The adult red swamp crayfish exhibits cyclic dimorphism, alternating between sexually active and inactive periods, and in the wild typically does not live longer than two to five years (GISD 2011, Huner and Barr 1991, Smart et al. 2002).

The red swamp crayfish exhibits two types of behaviors—one a wandering phase which involves short peaks of high speed of movement, the other an immobile stage during which it hides in its burrow by day and only comes out at dusk to forage. Breeding male crayfish in the wandering phase may travel as far as 17 km from their site of origin within four days (GISD 2011). Nocturnal activity in the stationary phase does not appear to be driven by predatory avoidance (many of red swamp crayfish predators are also nocturnal) or prey capture (mostly herbivorous; Gherardi et al. 2000).

Means of Introduction: Nonnative populations in the United States are likely to have resulted as a release from aquaculture or from the aquarium trade (Simon and Thoma 2006, Thoma and Jezerinac 2000; Kilian et al. 2012). This species' striking red color has lead to commercial advertisement as freshwater "lobster" for aquariums and may have sped up the species' advance on the west branch of the Grand Calumet River in Indiana and Illinois (Simon et al. 2005).

The red swamp crayfish is readily available though the biological supply trade and may be released following classroom or laboratory use (Larson and Olden 2008; Kilian et al. 2012). It is also popular among anglers as bait for largemouth bass (WDFW 2003). Intended disposal via the sanitary system (being flushed down toilets) is likely to be ineffective, as many *P. clarkii* has been seen in urban zones around waste water treatment areas, having apparently survived treatment (Indiana Biological Survey 2008).

The Sandusky Bay, OH populations likely stem from an attempted introduction to see if they could get a harvestable population established for human consumption (R. Thoma, Midwest Biodiversity Institute, pers. comm.). This species is commercially cultured in the southern U.S., particularly in Louisiana, where industry profits exceed $150 million annually and the fishery is an integral part of the state's culture and economy (McAlain and Romaire 2011). Alternately, there is a remote chance these red swamp crayfish were introduced from infested Ohio State Fish Hatcheries during a fish stocking event (R. Thoma, Midwest Biodiversity Institute, pers. comm.).

Status: Established in coastal waters of Lake Erie and Lake Michigan.

Impact of Introduction:

Summary of species impacts derived from literature review. Click on an icon to find out more...

Ecological	Economic	Human Health

Predation/Herbivory:

Procambarus clarkii has been shown to reduce macrophyte density through feeding, and to reduce macrophyte diversity through selective consumption (Cronin et al. 2002; Smart et al. 2002). The species also feeds on a variety of biota, including waterfleas, insects, and other crayfish (Gutiérrez- Yurrita et al. 1998). It's consumption of the egg masses and juveniles of the threatened California newt, *Taricha torosa*, the endangered razorback sucker *Xyrauchen texanus*, and the native toad *Bufo calamita*, is a cause for concern for the species (Gamradt and Kats 1996; Mueller et al. 2006; Cruz et al. 2006).

Competition:

Procambarus clarkii outcompetes not only other native crayfish (Mueller et al. 2006; Gheraridi and Daniels 2004), but also other native animals, such as dragonfly nymphs (Bucciarelli et al 2018), and amphibians (Bélouard et al 2019), reducing their density in the habitat.

Habitat Alteration:

Procambarus clarkii can be a vector for spreading other nonindigenous species or transplanting natives outside of their range via commensalism or through their gut (Clark and Wroten 1978; Lovas-Kiss et al. 2018). The species builds extensive burrows along shorelines that collapse and create erosion (Barbaresi et al. 2004).

Parasites:

Procambarus clarkii introduces novel parasites that are shown to be of high impact, in that the parasite results in high mortality of its hosts (Mastitsky et al 2010).

Agriculture and Aquaculture:

Procambarus clarkii damages wet-seeded rice fields in California and other areas outside of its range (Anastácio et al.1997; Anastácio et al. 2000; Anastácio et al. 2005).

Infrastructure:

The burrowing activity of Procombarus clarkii damages levees, dams, and water control devices (Correia and Ferreira 1995).

Remarks: Michigan Department of Natural Resources discovered in July 2015 that anglers are purchasing red swamp crayfish from food markets and using them as live bait. Several dead, red swamp crayfish were found near a popular fishing site in Lake Macatawa in Ottawa County, Michigan. To respond to this discovery, Michigan DNR will set traps and seines in Lake Macatawa.

A population of crayfish originally identified as *Procambarus clarkii* from the Seneca system, **New York** was later verifed as *Procambarus acutus* (11/28/2017).

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Other Resources:
A Brief Guide to Crayfish Identification in the Pacific Northwest

Global Invasive Species Database (GISD)

Minnesota Sea Grant

National Exotic Marine and Estuarine Species Information System (NEMESIS)

Wisconsin DNR

Minnesota DNR

NatureServe

Animal Diversity Web

National Park Service Aquatic Nuisance Species Pocket Guide

Great Lakes Waterlife

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